CLASS 9 · PART TEST-1 · STUDENT PERFORMANCE REPORT

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Campus HYD/Nallakunta DD Colony
JEE Mains 08 Oct 2026
JEE Advanced 09 Oct 2026

JEE Mains

08 Oct 2026 · 74 questions

144 / 296 — **48.6%**

214	145	15/79	37
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **154.4 / 300** (51.5%), 5 of 6 tests → this test **48.6% ▼ 2.8**

152 marks lost: 140 on 28 wrong answers (incl. 28 negative) · 12 on 3 skipped

JEE Advanced

09 Oct 2026 · 50 questions

92 / 176 — **52.3%**

71	51	6/78	9†
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **83.7 / 180** (46.5%), 4 of 6 tests → this test **52.3% ▲ 5.8**

84 marks lost: 79 on 21 wrong answers (incl. 7 negative) · 0 on 0 skipped · 5 on partly-correct answers

1 Subject scorecard

Your marks in each subject, against the 79 students of this program on campus and against your own weekly-test average.

SUBJECT	MARKS	SCORE VS CAMPUS	SUBJECT AIR	CAMPUS AVG / BEST	WEEKLY AVG	CHANGE	RIGHT / WRONG / SKIP
JEE Mains 144 / 296 · 48.6%							
Maths	60/100	60%	209	43.1 / 95	74.2/100 74.2%	▼ 14.2	17 / 8 / 0
Physics	38/100	38%	254	21.9 / 90	49.2/100 49.2%	▼ 11.2	12 / 10 / 3
Chemistry	46/96	48%	400	41.4 / 81	30.6/100 30.6%	▲ 17.3	14 / 10 / 0
JEE Advanced 92 / 176 · 52.3%							
Maths	46/60	77%	45	24.5 / 60	35.9/60 59.8%	▲ 16.9	14 / 3 / 0
Physics	25/60	42%	132	12.3 / 56	27.1/60 45.1%	▼ 3.5	8 / 9 / 0
Chemistry	21/56	38%	341	15.8 / 37	20.8/60 34.6%	▲ 2.9	7 / 9 / 0

Bar = your score % | campus average | best on campus Change = this test % − weekly-test average %

What this report says

1. **Maths** is your strongest JEE Mains subject: 60/100 (60%), All-India subject rank 209.
2. **Physics** needs the most work in JEE Mains: 38/100 (38%), subject rank 254; you lost 13 of its 25 questions.
3. **Maths** is your strongest JEE Advanced subject: 46/60 (77%), All-India subject rank 45.
4. **Chemistry** needs the most work in JEE Advanced: 21/56 (38%), subject rank 341; you lost 9 of its 16 questions.
5. Biggest fall against your weekly-test average: **JEE Mains Maths** – 60% here against 74% on average.
6. Biggest improvement: **JEE Mains Chemistry** – 48% here against 31% on average.
7. **5 questions** you lost were answered correctly by 70%+ of the batch's top 100 – about **24 marks** within easy reach (Mains Q14, Q23, Q46, Q65 · Advanced Q21).
8. Negative marking cost you **28 in JEE Mains and 7 in JEE Advanced**.
9. You attempted almost everything (only 3 questions skipped across both papers), so the marks to gain are in **accuracy**: wrong answers – 28 in JEE Mains and 21 in JEE Advanced.

JEE Mains – question by question

144 / 296 · 48.6%
08 Oct 2026

2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	**1**	2	3	4	5	**6**	7	**8**	9	10	11	12	**13**	**14**	15	16	**17**	18	19	**20**	21	22	**23**	24	25
Physics	**26**	27	**28**	29	**30**	**31**	32	33	34	**35**	36	37	**38**	39	40	41	**42**	43	**44**	**45**	**46**	**47**	**48**	49	**50**
Chemistry	51	52	53	54	55	56	57	~~58~~	**59**	60	61	**62**	**63**	64	**65**	**66**	**67**	68	69	70	**71**	72	**73**	**74**	**75**

3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 79 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

8 of 25 questions lost · **40** marks · ● 2 easy miss · ● 3 within reach · ● 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q14	Easy miss	**Trigonometric Equations**	Single correct · +4 / −1	**Wrong**	5	**84%**	58%
Q23	Easy miss	**Inverse Trigonometric Functions**	Numerical · +4 / −1	**Wrong**	5	**71%**	48%
Q13	Within reach	**Trigonometric Equations** – Solutions of tan x + sec x = 2 cos x in [0, 2π]	Single correct · +4 / −1	**Wrong**	5	**66%**	58%
Q6	Within reach	**Inverse Trigonometric Functions**	Single correct · +4 / −1	**Wrong**	5	**58%**	35%
Q1	Within reach	**Functions**	Single correct · +4 / −1	**Wrong**	5	**54%**	41%
Q8	Tough one	**Functions**	Single correct · +4 / −1	**Wrong**	5	**37%**	27%
Q17	Tough one	**Inverse Trigonometric Functions**	Single correct · +4 / −1	**Wrong**	5	**34%**	35%
Q20	Tough one	**Inverse Trigonometric Functions** – Composite inverse-trig function – counting solutions	Single correct · +4 / −1	**Wrong**	5	**19%**	13%

Physics

13 of 25 questions lost · **62** marks · ● 1 easy miss · ● 4 within reach · ● 8 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q46	Easy miss	**Circular Motion**	Numerical · +4 / −1	**Wrong**	5	**81%**	57%
Q50	Within reach	**Momentum & Collisions**	Numerical · +4 / −1	**Wrong**	5	**60%**	41%
Q44	Within reach	**Circular Motion**	Single correct · +4 / −1	**Wrong**	5	**50%**	41%
Q48	Within reach	**Momentum & Collisions**	Numerical · +4 / −1	**Wrong**	5	**47%**	34%
Q30	Within reach	**Circular Motion**	Single correct · +4 / −1	**Skipped**	4	**43%**	27%
Q47	Tough one	**Laws of Motion**	Numerical · +4 / −1	**Skipped**	4	**36%**	24%
Q35	Tough one	**Circular Motion** – Power delivered by the centripetal force	Single correct · +4 / −1	**Wrong**	5	**34%**	18%
Q31	Tough one	**Work, Energy & Power** – KE vs height for a stone thrown up from a cliff	Single correct · +4 / −1	**Wrong**	5	**32%**	10%
Q42	Tough one	**Work, Energy & Power** – Change in PE of a ball thrown from a tower	Single correct · +4 / −1	**Skipped**	4	**32%**	15%
Q26	Tough one	**Momentum & Collisions** – KE loss in head-on collision from relative velocities	Single correct · +4 / −1	**Wrong**	5	**31%**	22%
Q38	Tough one	**Work, Energy & Power** – Work along different paths; conservative or not	Single correct · +4 / −1	**Wrong**	5	**25%**	13%
Q28	Tough one	**Friction** – Friction vs incline angle graph (angle of repose)	Single correct · +4 / −1	**Wrong**	5	**21%**	11%
Q45	Tough one	**Work, Energy & Power** – Speed on hitting the ground – energy conservation	Single correct · +4 / −1	**Wrong**	5	**18%**	6%

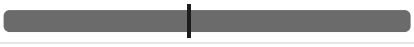
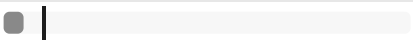
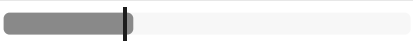
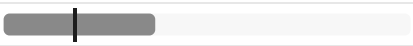
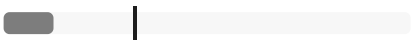
Chemistry

10 of 24 questions lost · **50** marks · ● 1 easy miss · ● 3 within reach · ● 6 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q65	Easy miss	**General Organic Chemistry**	Single correct · +4 / −1	**Wrong**	5	**70%**	62%
Q62	Within reach	**General Organic Chemistry**	Single correct · +4 / −1	**Wrong**	5	**66%**	47%
Q66	Within reach	**General Organic Chemistry** – False order of −I effect	Single correct · +4 / −1	**Wrong**	5	**45%**	39%
Q59	Within reach	**Isomerism**	Single correct · +4 / −1	**Wrong**	5	**43%**	38%
Q73	Tough one	**General Organic Chemistry**	Numerical · +4 / −1	**Wrong**	5	**32%**	23%
Q67	Tough one	**General Organic Chemistry** – Structure and hybridisation of the methyl carbocation	Single correct · +4 / −1	**Wrong**	5	**26%**	28%
Q74	Tough one	**General Organic Chemistry** – Compounds giving a positive bicarbonate test	Numerical · +4 / −1	**Wrong**	5	**20%**	11%
Q63	Tough one	**Hydrocarbons** – Monochlorination products incl. stereoisomers	Single correct · +4 / −1	**Wrong**	5	**16%**	20%
Q75	Tough one	**General Organic Chemistry** – Bases stronger than aniline	Numerical · +4 / −1	**Wrong**	5	**11%**	9%
Q71	Tough one	**Isomerism** – Count of optically active compounds	Numerical · +4 / −1	**Wrong**	5	**6%**	10%

4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Inverse Trigonometric Functions	10	**20**/40		48%	**Q6**, **Q17**, **Q20**, **Q23**
Functions	7	**18**/28		35%	**Q1**, **Q8**
Trigonometric Equations	7	**18**/28		44%	**Q13**, **Q14**
Sequences & Series	1	**4**/4		45%	– none
Physics					
Momentum & Collisions	3	**-3**/12		25%	**Q26**, **Q48**, **Q50**
Work, Energy & Power	5	**1**/20		9%	**Q31**, **Q38**, **Q42**, **Q45**
Circular Motion	7	**9**/28		30%	**Q30**, **Q35**, **Q44**, **Q46**
Friction	2	**3**/8		17%	**Q28**
Laws of Motion	8	**28**/32		23%	**Q47**
Chemistry					
General Organic Chemistry	10	**5**/40		32%	**Q62**, **Q65**, **Q66**, **Q67**, **Q73**, **Q74**, **Q75**
Hydrocarbons	4	**11**/16		42%	**Q63**
Isomerism	10	**30**/40		55%	**Q59**, **Q71**

Bar = your % of the chapter's marks | campus average

5 JEE Mains · Maths – questions to revisit

8 questions · easy misses first
cropped from the question paper

Q14 Trigonometric Equations

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **84%** got it right · Campus: **58%** (46 of 79)

Easy miss
You: Wrong

14. If $2\sin^2 x + 3\sin x - 2 > 0$ and $x^2 - x - 2 < 0$ then $x \in ?$

A) $\left(\frac{\pi}{6}, 2\right)$ B) $\left(-1, \frac{5\pi}{6}\right)$ C) $\left(-1, \frac{\pi}{6}\right)$ D) $\left(2, \frac{5\pi}{6}\right)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q23 Inverse Trigonometric Functions

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5
Top 100: **71%** got it right · Campus: **48%** (38 of 79)

Easy miss
You: Wrong

23. $\cos^2\left(\tan^{-1}\left(\sin\left(\cot^{-1}(3)\right)\right)\right) = ...$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13 Trigonometric Equations – Solutions of tan x + sec x = 2 cos x in [0, 2π]

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **66%** got it right · Campus: **58%** (46 of 79)

Within reach
You: Wrong

13. Number of solution(s) of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$ is

A) 1 B) 2 C) 3 D) 4

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q6

Inverse Trigonometric Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **58%** got it right · Campus: **35%** (28 of 79)

Within reach

You: Wrong

6. $\tan^{-1}(2)+\tan^{-1}(3)=\ldots\ldots\ldots\ldots$

A) $\left(-\frac{\pi}{4}\right)$ B) $\frac{\pi}{4}$ C) $\frac{3\pi}{4}$ D) $-\frac{3\pi}{4}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q1

Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **54%** got it right · Campus: **41%** (32 of 79)

Within reach

You: Wrong

1. The function $f: R-\{0\}\to R, f(x)=\frac{x}{e^x-1}+\frac{x}{2}+1$ is

A) even B) odd C) One- one D) periodic

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q8

Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **37%** got it right · Campus: **27%** (21 of 79)

Tough one

You: Wrong

8. Let $f(x)$ denote the square of sum of digits of $n\in N$. Let $f_2(x)=f(f(x)), f_3(x)=f(f(f(x)))$ and so on. Then, $\frac{f_{2026}(2011)-f_{2023}(2011)}{f_{2024}(2011)-f_{2025}(2011)}=\ldots\ldots\ldots\ldots$

A) 0 B) -1 C) 1 D) 2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17

Inverse Trigonometric Functions
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **34%** got it right · Campus: **35%** (28 of 79)

Tough one
You: Wrong

17. Let $\left(\sin^{-1}x\right)^2+\left(\cos^{-1}x\right)^2=k\left(where\ \ x\in[-1,1]\right)$ then k cannot be

A) $\frac{\pi^2}{8}$ B) $\frac{\pi^2}{3}$ C) $\frac{\pi^2}{9}$ D) π^2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20

Inverse Trigonometric Functions – Composite inverse-trig function – counting solutions
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **19%** got it right · Campus: **13%** (10 of 79)

Tough one
You: Wrong

20. Consider the functions $rage(x)=\sin^{-1}\sqrt{x}+e^{\left(\tan^{-1}\left(\sqrt{x-1}\right)\right)^2}+\left(\cos^{-1}\left(\sqrt{x}\right)\right)^3$ and $bait(x)=x^2+x+1$ within their respective domains. Then the number of solutions of $rage\left(bait(x)\right)<3$ is/are ……..

A) 0 B)1 C)2 D)3

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Physics – questions to revisit

13 questions · easy misses first
cropped from the question paper

Q46 — Circular Motion

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 5
Top 100: **81%** got it right · Campus: **57%** (45 of 79)

Easy miss
You: Wrong

46. A mass 5kg is moving along a circular path of radius 1m. if mass moves with 300rpm, its kinetic energy would be ..... π^2J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q50 — Momentum & Collisions

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 5
Top 100: **60%** got it right · Campus: **41%** (32 of 79)

Within reach
You: Wrong

50. When kinetic energy of a body becomes 64 times of its original value, the percentage increase in momentum of body will be .....%

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q44 — Circular Motion

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **50%** got it right · Campus: **41%** (32 of 79)

Within reach
You: Wrong

44. A wheel is subjected to uniform angular acceleration about its axis. Initially angular velocity is zero. In the first 2sec, it rotates through an angle θ_1. In the next 2sec, it rotates through an additional angle θ_2. The ratio of $\frac{\theta_2}{\theta_1}$ is

A) 1 B) 3 C) 2 D) 5

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q48

Momentum & Collisions

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 5

Top 100: **47%** got it right · Campus: **34%** (27 of 79)

Within reach

You: Wrong

48. An artillery piece of mass 10kg fires a shell of mass 2kg horizontally. Instantaneously after the firing, the ratio of kinetic energy of shell and that of artillery is …….

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Circular Motion

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **43%** got it right · Campus: **27%** (21 of 79)

Within reach

You: Skipped

30. When a bucket containing water is rotated fast in a vertical circle of radius R, the water in the bucket doesn't spill provided that bucket is

A) whirled around with maximum speed of $\sqrt{2gR}$

B) whirled around with minimum speed of $\sqrt{\frac{gR}{2}}$

C) having a rpm of $n=\sqrt{900g/\pi^2R}$

D) having a rpm of $n=\sqrt{3600g/\pi^2R}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47

Laws of Motion

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **36%** got it right · Campus: **24%** (19 of 79)

Tough one

You: Skipped

47. A body with mass 5kg is acted upon by a force $\vec{F}=\left(-3\hat{i}+4\hat{j}\right)N$. If its initial velocity at $t=0\text{sec}$ is $u=\left(6\hat{i}-12\hat{j}\right)m/s$. The time at which it will just have a velocity along y-axis ……sec

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35

Circular Motion – Power delivered by the centripetal force

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **34%** got it right · Campus: **18%** (14 of 79)

Tough one

You: Wrong

35. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration changing with time 't' as $a_c = k^2t^2$, where K is a constant. The power delivered to the particle by centripetal force acting on it is

A) zero B) mk^2rt C) mk^2r^2t D) $mk^2\sqrt{rt}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31

Work, Energy & Power – KE vs height for a stone thrown up from a cliff

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

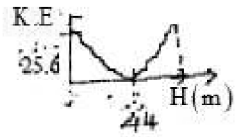
Top 100: **32%** got it right · Campus: **10%** (8 of 79)

Tough one

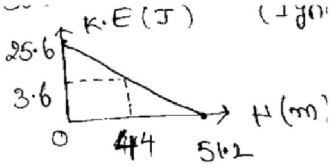
You: Wrong

31. A stone of mass 50g is thrown vertically upwards with a velocity of 12m/s from edge of a cliff 44m high. Identify kinetic energy of stone versus height from bottom of cliff $\left(g = 10m/s^2\right)$(ignore air resistance)

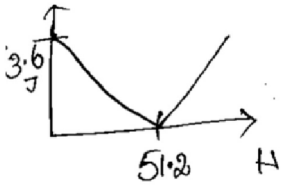
A)



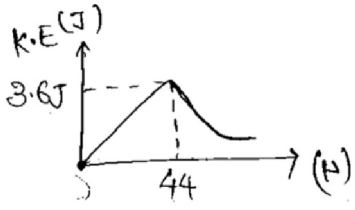
B)



C)



D)



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q42

Work, Energy & Power – Change in PE of a ball thrown from a tower

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **32%** got it right · Campus: **15%** (12 of 79)

Tough one

You: Skipped

42. At t =0 sec a 1kg ball is thrown from top to a tall tower with velocity $\vec{V} = \left(8\hat{i} + 9\hat{j}\right)$ m/s. The change in potential energy of ball – earth system between $t = 0$ *to* $t = 5\text{sec}$ is

$\left(g = 10m/s^2\right)$

A) 800J B) 1536J C) 2100J D) -800J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q26

Momentum & Collisions – KE loss in head-on collision from relative velocities

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **31%** got it right · Campus: **22%** (17 of 79)

Tough one

You: Wrong

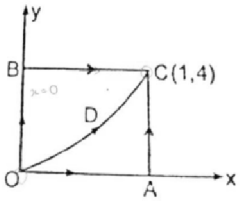
26. Two bodies of masses 3 kg and 2 kg collide head on. Their relative velocities before and after collision are 15m/s and 5 m/s respectively. The loss of kinetic energy of system is

A) 120J B) 100J C) 80J D) 240J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

38. A particle is moved along the different paths OAC, OBC and ODC as shown in the fig. path ODC is parabola, $y = 4x^2$. The work done by a force $\vec{F} = xy\hat{i} + x^2 y\hat{j}$ on the particle along



Column – I	**Column – II**
a)Path OAC	P) 2J
b) Path OBC	Q) $\frac{19}{3}$J
c) Path ODC	R) Conservative
d) Force is	S) 8J
	T) Non- conservative

A) $a \to Q, b \to S, c \to P, d \to T$ B) $a \to S, b \to P, c \to Q, d \to T$

C) $a \to S, b \to Q, c \to P, d \to T$ D) $a \to S, b \to Q, c \to S, d \to T$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28

Friction – Friction vs incline angle graph (angle of repose)

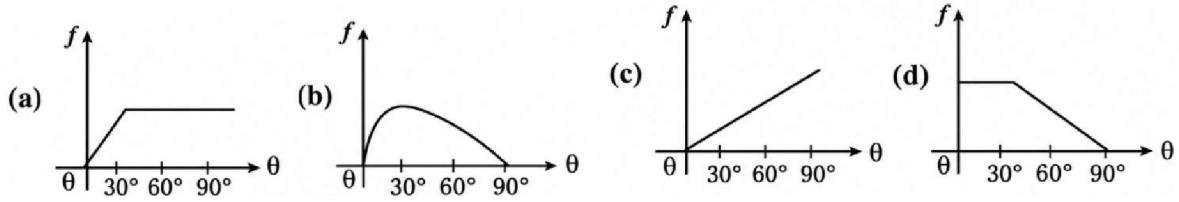
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **21%** got it right · Campus: **11%** (9 of 79)

Tough one

You: Wrong

28. A block is placed on a rough plane whose inclination to the horizontal (θ) can be varied. The angle of repose is 30°. The graph that correctly indicate the variation of frictional force (f) between block and the plane with θ is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q45

Work, Energy & Power – Speed on hitting the ground – energy conservation

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **18%** got it right · Campus: **6%** (5 of 79)

Tough one

You: Wrong

45. A cricket bowler releases the ball in two different ways

i) giving it only horizontal velocity and

ii) giving it horizontal velocity and small downward velocity

The speeds at the time release is same. Both are released at a height H from ground then which one will have greater speed when ball hits the ground?(Neglect air resistance)

A) Ist way B) IInd way C) same in two ways D) None of the above

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Chemistry – questions to revisit

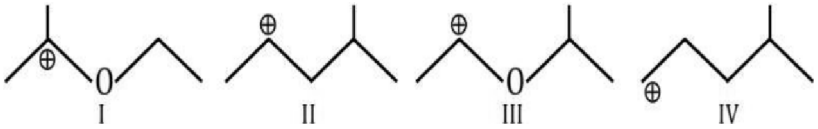
10 questions · easy misses first
cropped from the question paper

Q65

General Organic Chemistry
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **70%** got it right · Campus: **62%** (49 of 79)

Easy miss
You: Wrong

65. The correct stability order of the following species is :



A) II > IV > I > III

B) I > II > III > IV

C) II > I> IV > III

D) I > III > II > IV

Why did I lose this question? (write the reason and the correct approach)

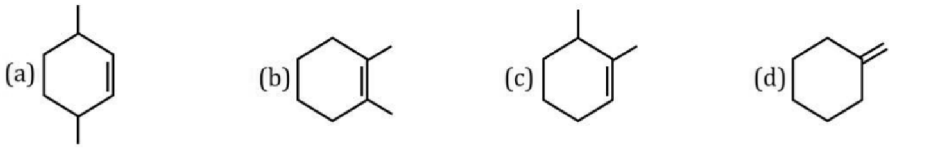
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q62

General Organic Chemistry
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **66%** got it right · Campus: **47%** (37 of 79)

Within reach
You: Wrong

62. Decreasing order of heat evolved upon catalytic hydrogenation of given reaction with a H_2 / Pd is (only based on hyper conjugation effect)



(A) b > c > a > d

(B) a > d > c > b

(C) d > c > a > b

(D) c > b > c > d

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q66

General Organic Chemistry – **False order of –I effect**

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **45%** got it right · Campus: **39%** (31 of 79)

Within reach

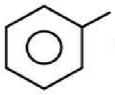
You: Wrong

66. Which of the following is false order of – I effect ?

A) $-F > -Cl > -Br > -I$

B) $-\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2$

C) $-F > -OH > -NH_2$

D)  $> -C \equiv CH > -H$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q59

Isomerism

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **43%** got it right · Campus: **38%** (30 of 79)

Within reach

You: Wrong

59. Which of the following statements regarding 1,2-dimethylcyclopropane (I) and 1,2-dimethyl cyclobutane (II) are wrong ?

A) Both of them show three stereoisomers each

B) The cis form of both is optically inactive (meso) and the trans form of both has a pair of enantiomers

C) The cis form of both has a pair of enantiomers and the trans form of both is optically inactive (meso)

D) The meso form of both is optically inactive due to the presence of plane of symmetry.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q73

General Organic Chemistry

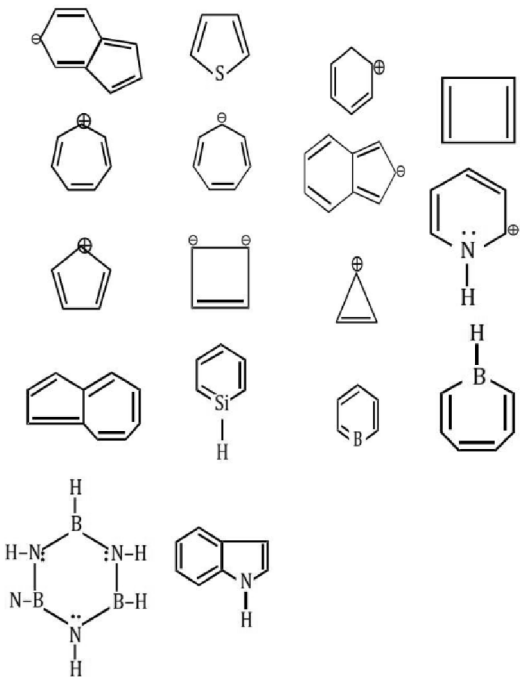
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **32%** got it right · Campus: **23%** (18 of 79)

Tough one

You: Wrong

73. How Many of the following are Aromatic compounds?



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q67

General Organic Chemistry – Structure and hybridisation of the methyl carbocation

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **26%** got it right · Campus: **28%** (22 of 79)

Tough one

You: Wrong

67. Wrong statement regarding methyl carbonium ion $\overset{\oplus}{C}H_3$

A) It is sp^2 hybridized

B) Vacant orbital is sp^2 hybridized

C) vacant orbital is perpendicular to molecular planar and is pure p-orbital

D) It is electrophilic with sextet of electron

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q74

General Organic Chemistry – Compounds giving a positive bicarbonate test

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **20%** got it right · Campus: **11%** (9 of 79)

Tough one

You: Wrong

74. How many of the following compounds can give positive bicarbonate test?

HCl, Picric Acid, Benzoic acid, 2,4-dintrophenol, 4-nitrophenol, Squaric Acid, formic Acid, Benzene Sulphuric acid, 1-Butyne, NH_4Cl .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q63

Hydrocarbons – Monochlorination products incl. stereoisomers

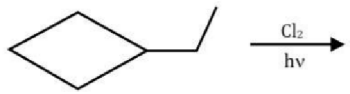
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **16%** got it right · Campus: **20%** (16 of 79)

Tough one

You: Wrong

63. Number of monochlorinated product when following compound undergo reaction with $Cl_2 / h\nu$ (including stereo isomers)



A) 5 B) 9 C) 10 D) 6

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q75

General Organic Chemistry – **Bases stronger than aniline**

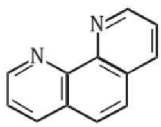
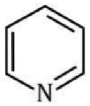
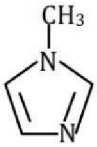
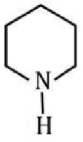
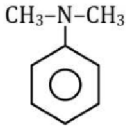
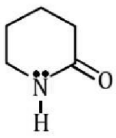
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **11%** got it right · Campus: **9%** (7 of 79)

Tough one

You: Wrong

75. How many of the following are more basic then Aniline?

NH_3, $PhCH_2NH_2$,  ,  ,  ,  ,  ,  , $CH_3—CH_2—NH_2$, $CH_3—C\equiv N$, 

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q71

Isomerism – **Count of optically active compounds**

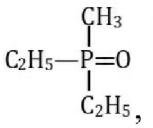
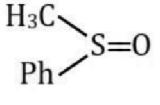
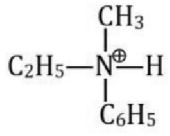
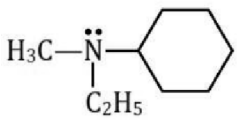
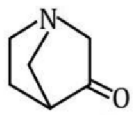
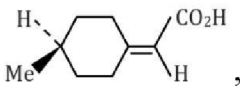
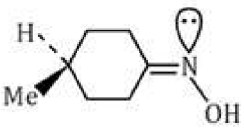
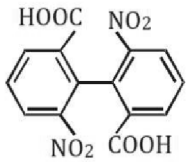
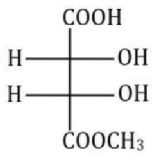
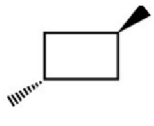
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **6%** got it right · Campus: **10%** (8 of 79)

Tough one

You: Wrong

71. Total no of compounds which are optically active are _____

 ,  ,  ,  ,  ,  ,  ,  ,  , 

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	**4**	5	6	7	**8**	9	10	11	12	13	14	15	16	**17**
Physics	**18**	19	**20**	**21**	22	23	24	25	26	**27**	**28**	29	**30**	31	**32**	**33**	**34**
Chemistry	**35**	36	37	38	39	40	**41**	42	**43**	**44**	~~45~~	**46**	**47**	**48**	49	**50**	**51**

7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 78 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

3 of 17 questions lost · **11** marks (+3 on partly-correct answers) · ● 1 within reach · ● 2 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q4	Within reach	**Functions**	Single correct · +3 / −1	Wrong	4	55%	65%
Q8	Tough one	**Functions** – Period from two lines of symmetry	Integer · +4 / 0	Wrong	4	12%	13%
Q17	Tough one	**Trigonometric Equations** – Sum of solutions of $\cos(\pi\sqrt{(x-100)})\cdot\cos(\pi\sqrt{x}) = 1$	Paragraph · +3 / 0	Wrong	3	6%	6%

Physics

9 of 17 questions lost · **33** marks (+2 on partly-correct answers) · ● 1 easy miss · ● 1 within reach · ● 7 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q21	Easy miss	**Work, Energy & Power**	Single correct · +3 / −1	Wrong	4	86%	76%
Q28	Within reach	**Momentum & Collisions**	Integer · +4 / 0	Wrong	4	49%	33%
Q20	Tough one	**Work, Energy & Power** – Maximum spring extension – energy method	Single correct · +3 / −1	Wrong	4	29%	27%
Q30	Tough one	**Friction** – Work by friction seen from an accelerating observer	Integer · +4 / 0	Wrong	4	19%	12%
Q18	Tough one	**Friction** – Static vs kinetic friction under a time-varying force	Single correct · +3 / −1	Wrong	4	16%	9%
Q34	Tough one	**Work, Energy & Power** – Work by conservative forces from the U–x graph	Paragraph · +3 / 0	Wrong	3	12%	9%
Q27	Tough one	**Friction** – Net work done by friction between sliding blocks	Integer · +4 / 0	Wrong	4	8%	10%
Q32	Tough one	**Friction** – Total force exerted by a block on an accelerating cube	Paragraph · +3 / 0	Wrong	3	8%	6%
Q33	Tough one	**Work, Energy & Power** – Work-energy theorem from a–t and U–x graphs	Paragraph · +3 / 0	Wrong	3	2%	1%

Chemistry

9 of 16 questions lost · **35** marks · ● 4 within reach · ● 5 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q35	Within reach	**General Organic Chemistry**	Single correct · +3 / −1	**Wrong**	**4**	**58%**	42%
Q44	Within reach	**General Organic Chemistry**	Integer · +4 / 0	**Wrong**	**4**	**56%**	46%
Q43	Within reach	**Hydrocarbons** – Alkane from O_2 needed; count secondary carbons	Integer · +4 / 0	**Wrong**	**4**	**47%**	28%
Q41	Within reach	**Isomerism**	Multiple correct · +4 / −2	**Wrong**	**6**	**43%**	42%
Q48	Tough one	**General Organic Chemistry** – Counting electrophiles	Paragraph · +3 / 0	**Wrong**	**3**	**37%**	27%
Q46	Tough one	**Isomerism** – Count of optically active compounds	Integer · +4 / 0	**Wrong**	**4**	**15%**	15%
Q51	Tough one	**Hydrocarbons** – Chiral products of free-radical monobromination	Paragraph · +3 / 0	**Wrong**	**3**	**15%**	4%
Q47	Tough one	**Hydrocarbons** – Dichlorination isomers of C_5H_{10} (VD 35)	Integer · +4 / 0	**Wrong**	**4**	**7%**	5%
Q50	Tough one	**Hydrocarbons** – Preparation of alkanes – count correct statements	Paragraph · +3 / 0	**Wrong**	**3**	**4%**	3%

8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Functions	6	≈ **14**/22		40%	**Q4**, **Q8**
Trigonometric Equations	5	≈ **14**/17		35%	**Q17**
Inverse Trigonometric Functions	6	≈ **21**/21		55%	– none
Physics					
Work, Energy & Power	4	≈ **-2**/12		22%	**Q20**, **Q21**, **Q33**, **Q34**
Momentum & Collisions	1	≈ **0**/4		33%	**Q28**
Friction	5	≈ **2**/17		12%	**Q18**, **Q27**, **Q30**, **Q32**
Circular Motion	4	≈ **15**/15		17%	– none
Laws of Motion	3	≈ **12**/12		39%	– none
Chemistry					
Isomerism	2	≈ **-2**/8		16%	**Q41**, **Q46**
Hydrocarbons	5	≈ **4**/18		22%	**Q43**, **Q47**, **Q50**, **Q51**
General Organic Chemistry	9	≈ **19**/30		43%	**Q35**, **Q44**, **Q48**

Bar = your % of the chapter's marks | campus average

9 JEE Advanced · Maths – questions to revisit

3 questions · easy misses first
cropped from the question paper

Q4 — Functions

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 4
Top 100: **55%** got it right · Campus: **65%** (51 of 78)

Within reach
You: Wrong

4. Consider the function $f:N\to N, f(n)=p$ where 'p' is the largest prime number that divides n^2+n+41. Then f is :

A) one-one B) many-one C)onto D) into

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q8 — Functions – Period from two lines of symmetry

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4
Top 100: **12%** got it right · Campus: **13%** (10 of 78)

Tough one
You: Wrong

8. Let $a\in R$. If $f(a-x)=f(a+x)\forall x\in R$, then $x=a$ is a line of symmetry in the graph of $y=f(x)$. Consider a non-constant function $g(x)$ such that $g(1+x)=g(1-x)\forall x\in R$ and $g(2026+x)=g(2026-x)\forall x\in R$. Then all such $g(x)$ must be periodic. The smallest positive number that could be a period for all such $g(x)$ is .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17 **Trigonometric Equations – Sum of solutions of cos(π√(x−100))·cos(π√x) = 1**

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **6%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Wrong

PARAGRAPH-II

Given positive integer 'k' , if we want to find positive integer 'n' and 'm' such that $n^2 - m^2 = k$ then we use $(n+m)(n-m) = k$ where we equate $n+m$ & $n-m$ to factors of k to find all possible values of (n, m).

Eg: $n^2 - m^2 = 6 \Rightarrow (n+m)(n-m) = (\pm 6)(\pm 1)$ or $(\pm 3)(\pm 2)$

17. The sum of real solution(s) of $\cos\left(\pi\sqrt{x-100}\right)\cos\left(\pi\sqrt{x}\right) = 1$ is .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Physics – questions to revisit

9 questions · easy misses first
cropped from the question paper

Q21

Work, Energy & Power
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4
Top 100: **86%** got it right · Campus: **76%** (59 of 78)

Easy miss
You: Wrong

21. A block of mass m is pulled by a constant power P placed on a rough horizontal plane. The friction coefficient between block and surface is μ. The maximum velocity of block is

A) $\frac{P}{mg}$ B) $\frac{\mu P}{mg}$ C) $\frac{P}{\mu mg}$ D) $\frac{P}{\mu^2 mg}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28

Momentum & Collisions
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **49%** got it right · Campus: **33%** (26 of 78)

Within reach
You: Wrong

28. Three objects A, B and C are kept in a straight line on a frictionless horizontal surfaces. These have masses m,2m and m respectively. The object A moves towards B with speed 9m/s and makes an elastic collision with it. There after B makes completely inelastic collision with 'C'. All motion occur on same straight line final velocity of object 'C' is .....m/s.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20

Work, Energy & Power – **Maximum spring extension – energy method**

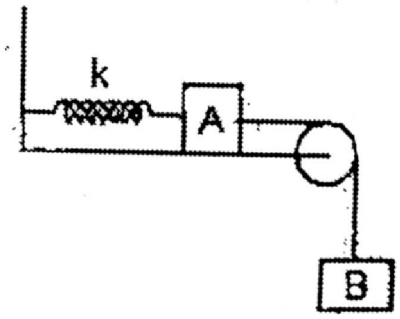
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **29%** got it right · Campus: **27%** (21 of 78)

Tough one

You: Wrong

20. All the surfaces are smooth system is released from rest with spring unstretched. Then maximum extension in spring will be $\left(m_A = m; m_B = 2m\right)$



A) $\frac{mg}{K}$ B) $\frac{2mg}{K}$ C) $\frac{3mg}{K}$ D) $\frac{4mg}{K}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Friction – **Work by friction seen from an accelerating observer**

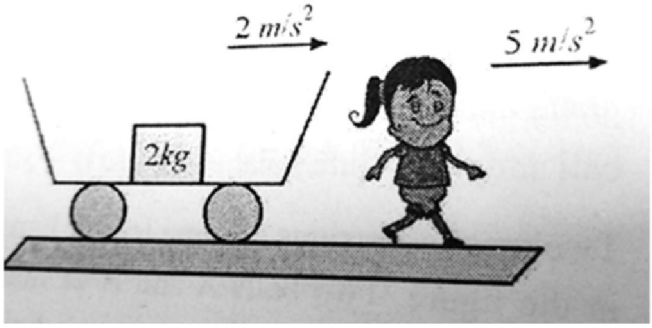
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **19%** got it right · Campus: **12%** (9 of 78)

Tough one

You: Wrong

30. An observer and a vehicle, both starts moving together from rest with acceleration $5m/s^2$ and $2m/s^2$ respectively. There is a 2 kg block on the floor of the vehicle and $\mu = 0.3$ between their surfaces. The work done by the frictional force on 2kg block as observed by running observer , during first 2seconds of motion is $\underline{-x}$ (J) $\left(g = 10m/s^2\right)$ find 'x'



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q18

Friction – **Static vs kinetic friction under a time-varying force**

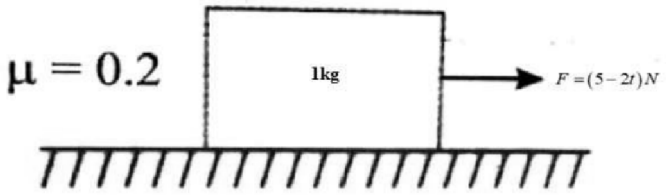
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **16%** got it right · Campus: **9%** (7 of 78)

Tough one

You: Wrong

18. The force acting on the block of mass 1 kg is given by $F = 5 - 2t$. The frictional force acting on the block after time $t = 2\text{sec}$ will be $\left(\mu = 0.2, g = 10m/s^2\right)$



A) 2N B) 3N C) 1N D) zero

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34

Work, Energy & Power – **Work by conservative forces from the U-x graph**

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **12%** got it right · Campus: **9%** (7 of 78)

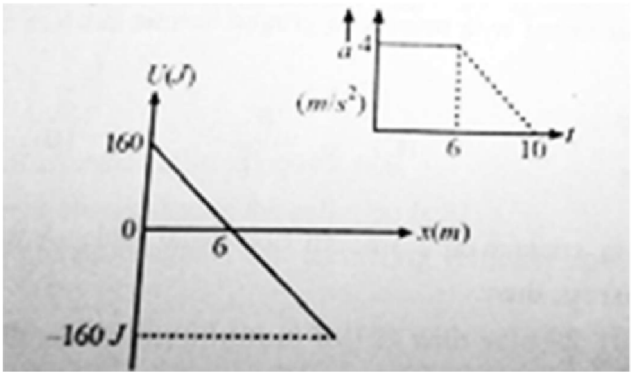
Tough one

You: Wrong

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



34. Work done by conservative forces _____ J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Friction – Net work done by friction between sliding blocks

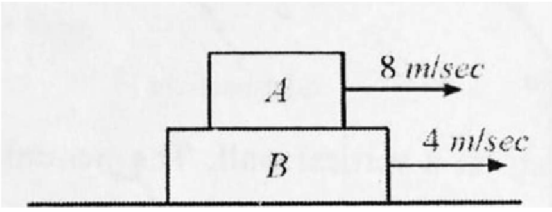
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **8%** got it right · Campus: **10%** (8 of 78)

Tough one

You: Wrong

27. Block A of mass 1kg is placed on rough surface of block B of mass 3kg. Block B is placed on smooth horizontal surface. Blocks are given velocities as shown. If net workdone by frictional force is "-n" J. value of n is ……..



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32

Friction – Total force exerted by a block on an accelerating cube

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

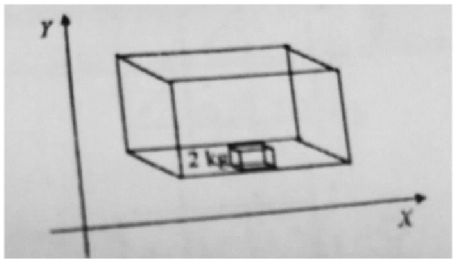
Top 100: **8%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Wrong

PARAGRAPH-I

A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V}=\left(5t\hat{i}+2t\hat{j}\right)m/s$. Here 't' is second. The block is in rest w.r.t cube and $\mu=0.5$ between block and surface of cube. $\left(g=10m/s^2\right)$



32. Total force exerted by block on cube is _____ N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

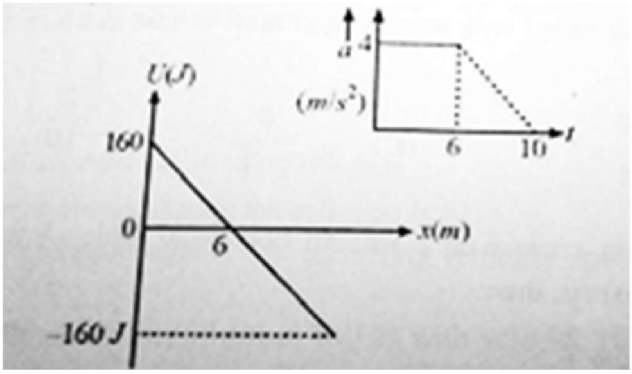
Q33 **Work, Energy & Power** – **Work-energy theorem from a-t and U-x graphs**
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **2%** got it right · Campus: **1%** (1 of 78)

Tough one
You: Wrong

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



33. Work done by all forces _____ J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Chemistry – questions to revisit

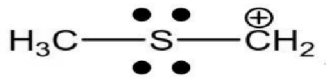
9 questions · easy misses first
cropped from the question paper

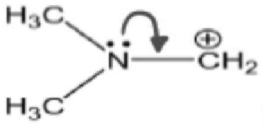
Q35 General Organic Chemistry

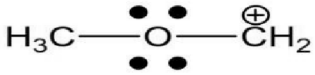
JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4
Top 100: **58%** got it right · Campus: **42%** (33 of 78)

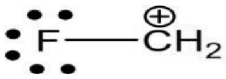
Within reach
You: Wrong

35. Which of the following cation would have greatest stability?

A) 

B) 

C) 

D) 

Why did I lose this question? (write the reason and the correct approach)

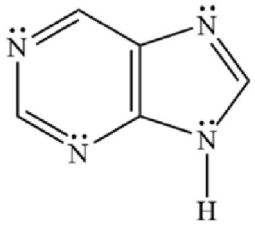
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q44 General Organic Chemistry

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4
Top 100: **56%** got it right · Campus: **46%** (36 of 78)

Within reach
You: Wrong

44. The purine heterocycle occurs mainly in the structure of DNA. Identify number of 'N' atoms having localized lone pair of electrons.



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q43

Hydrocarbons – Alkane from O_2 needed; count secondary carbons

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **47%** got it right · Campus: **28%** (22 of 78)

Within reach

You: Wrong

43. one mole of alkane (Y) requires 8 moles of oxygen for complete combustion on chlorination with $Cl_2 / h\gamma$, gives four monochlorinated products (including stereo). The total number of secondary carbon atoms per each (Y) is ……..

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q41

Isomerism

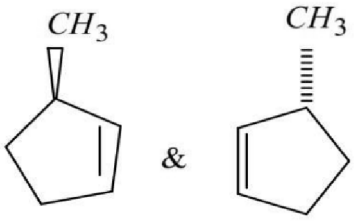
JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 6

Top 100: **43%** got it right · Campus: **42%** (33 of 78)

Within reach

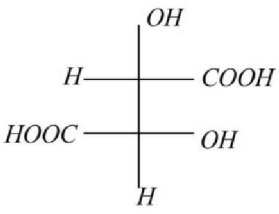
You: Wrong

41. Select the correct statements:



A) are enantiomers

B) The Boiling point or E-isomers is more than Z-isomers of 1-chloro propene



C) is optically active & may be dextro rotatory compound

D) Molecular formula C_3H_6O, possible to show geometrical isomerism.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q48

General Organic Chemistry – **Counting electrophiles**

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **37%** got it right · Campus: **27%** (21 of 78)

Tough one

You: Wrong

PARAGRAPH-I

Electrophiles and nucleophiles are fundamental concepts of organic chemistry playing crucial roles in various reactions. Electron loving species that accept electron pairs to form covalent bonds are electrophiles, Nucleus loving species that donate electron pair to form covalent bonds.

48. How many of the following are electrophiles

A) Br^+ B) Na^+ C) CN^- D) BF_3

E) $BeCl_2$ F) NH_2^- G) SH^- H) OR^-

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q46

Isomerism – **Count of optically active compounds**

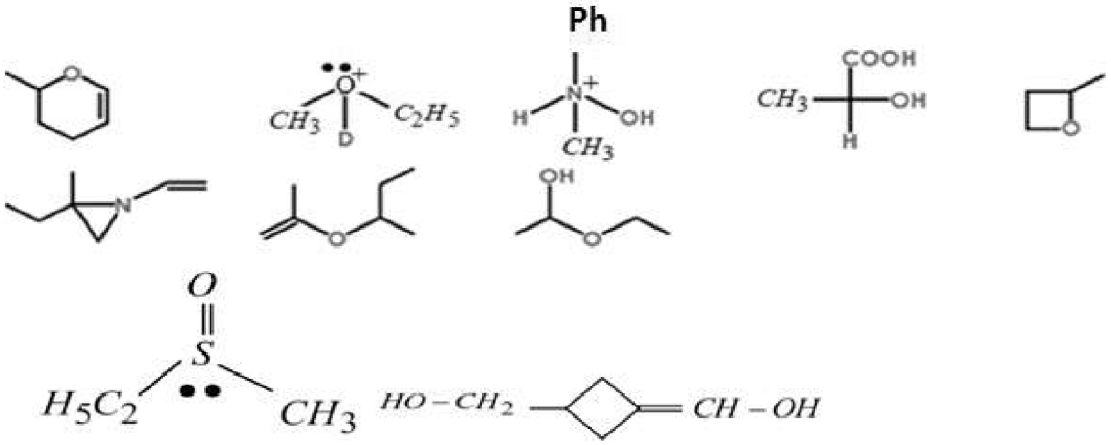
JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **15%** got it right · Campus: **15%** (12 of 78)

Tough one

You: Wrong

46. How many given compounds are optically active



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q51

Hydrocarbons – Chiral products of free-radical monobromination

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **4%** (3 of 78)

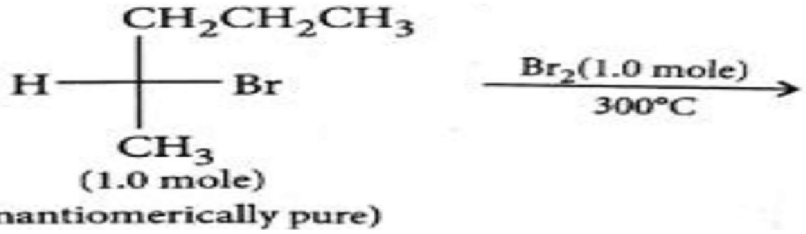
Tough one

You: Wrong

PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

51. In the following monobromination reaction, the number of possible chiral products is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47

Hydrocarbons – Dichlorination isomers of C_5H_{10} (VD 35)

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **7%** got it right · Campus: **5%** (4 of 78)

Tough one

You: Wrong

47. A hydrocarbon [X] is having vapour density 35 and it forms only 1 monochloro product under sunlight. How many isomeric products (including stereo) are possible when [X] undergoes dichlorination?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

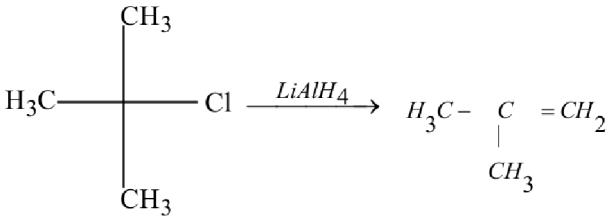
PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

50. Find the No of correct statements out of the following

A) Alkanes are prepared by catalytic hydrogenation of Alkenes where both hydrogen atoms add to the same face of the alkene (syn-addition).

B) Alkyl halides when reacts with Lithium Aluminium Hydride form corresponding alkane: $R-X \xrightarrow{LiAlH_4} R-H$ for R : methyl , $1^0, 2^0$

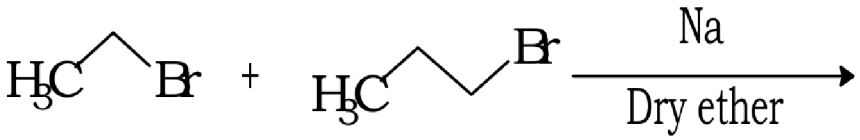


C)In case of Tertiary Alkyl halide is observed.

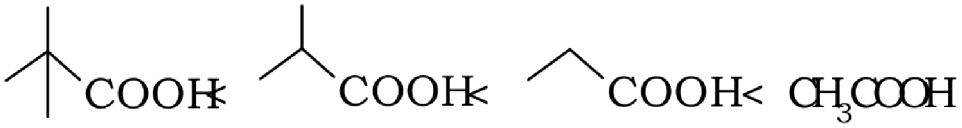
D) Grignard's reagent when treated with a polar protic solvent gives alkane:

$R-Mg+HA \rightarrow R-H+MgX(A)$

E) The different compounds obtained by the following reaction are 7 (only organic compounds)



F) Rate of sodalime decarboxylation for the following acids follow:



G) The side products of potassium butanoate during Kolbe's Electrolysis are propane, propene and propyl butanoate

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

10 How to read this report

Where the numbers come from. Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q58 — JEE Advanced: Q45. **Program rank** = rank in Pragya G9 (EXT) Achievers across all campuses, from the batch's question-wise sheet († JEE Advanced: that sheet was graded before the final key, so this rank may differ slightly). **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. In JEE Advanced, 5 marks went on multiple-correct answers that were only partly correct. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.